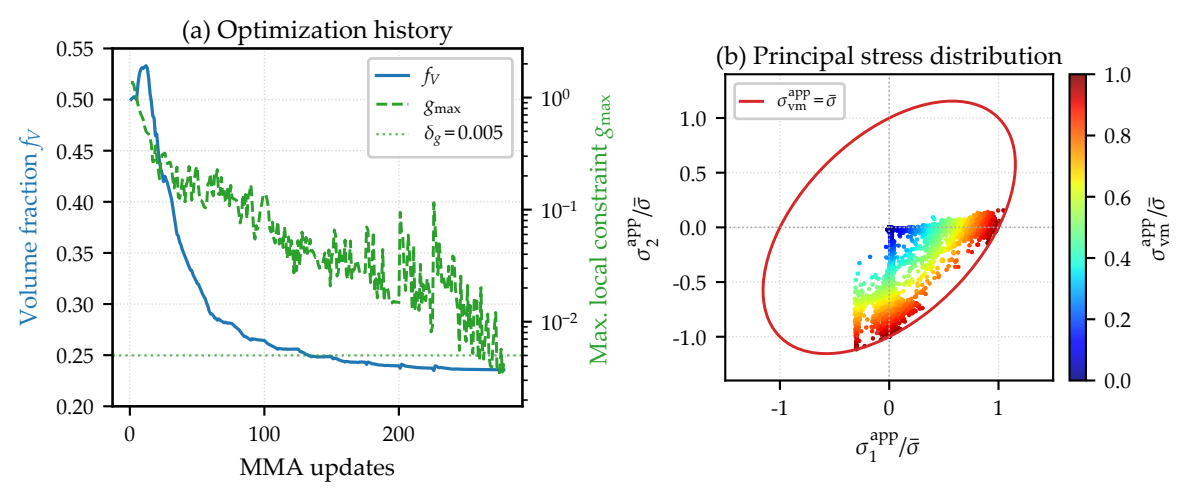

(a) Optimization history
Volume fraction f_V
Max. local constraint $g_{\max}$
MMA updates
f_V
$g_{\max}$
$\delta_g = 0.005$
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0
100
200
10^0
10^{-1}
10^{-2}
(b) Principal stress distribution
$\sigma_{\mathrm{vm}}^{\mathrm{app}} = \bar{\sigma}$
$\sigma_2^{\mathrm{app}}/\bar{\sigma}$
$\sigma_1^{\mathrm{app}}/\bar{\sigma}$
$\sigma_{\mathrm{vm}}^{\mathrm{app}}/\bar{\sigma}$
1.0
0.5
0.0
-0.5
-1.0
-1
0
1
0.8
0.6
0.4
0.2
0.0